Web Development Track • Phase 2 • Lecture 10

Modern Layouts: Flexbox Deep Dive

Effortless 1D layouts, parent container controls, child growth dynamics, and the ultimate centering trick.

Main vs Cross Axes

Parent vs Child Rules

Grow, Shrink & Basis

Perfect Centering

LAYOUT FOUNDATIONS

The Mental Model: Container & Items

Flex Container (Parent)

The conductor managing the group.

Activation: Initiated via display: flex

Direction: Controls row or column flow

Alignment: Spaces out and centers items

```
.parent { display: flex; }
```

Flex Items (Children)

The immediate tags inside the container.

Direct Children: Grandchildren are not items

Flexibility: Expand or shrink into space

Self Alignment: Can override parent rules

```
.child { flex-grow: 1; }
```

CORE AXIS PRINCIPLE

The Main Axis vs. Cross Axis

PRIMARY FLOW

The Main Axis

Defined directly by flex-direction.

```
/* Horizontal main axis (Default) */ flex-direction: row; /* Controlled via justify-content */
```

Left-to-right by default in LTR languages.

PERPENDICULAR FLOW

The Cross Axis

Always runs at 90 degrees to main axis.

```
/* Vertical cross axis when row */ align-items: center; /* Runs top-to-bottom */
```

Swaps automatically when direction becomes column.

CONTAINER PROPERTY

flex-direction: Setting the Flow

DEFAULT

row

Items side-by-side horizontally left to right.

VERTICAL

column

Items stacked vertically top to bottom.

REVERSED

row-reverse

Horizontal items starting from right edge.

INVERTED

column-reverse

Vertical items starting from bottom edge.

MAIN AXIS ALIGNMENT

justify-content: Distributing Space

space-between

First item at start, last item at end edge.

center

Packs all items together at the center point.

space-around / space-evenly

Equal margins or uniform gap across all slots.

```
/* Classic responsive navbar layout */ .navbar { display: flex; justify-content:
space-between; align-items: center; } /* Logo on far-left, links on far-right */
```

CROSS AXIS ALIGNMENT

align-items: Perpendicular Control

DEFAULT

stretch

Fills container cross-axis height automatically.

VERTICAL CENTER

center

Centers items vertically inside row container.

TOP / START

flex-start

Packs items flush against cross-axis origin.

BASELINE

baseline

Aligns based on text font baselines.

RESPONSIVE SPACING

flex-wrap & The gap Property

MULTI-LINE WRAPPING

flex-wrap: wrap;

Enables wrapping when items exceed container width.

nowrap (Default): Squeezes items onto one line

wrap: Breaks into next row gracefully

CLEAN SEPARATION

gap: 20px;

Sets space between items without margin hacks.

```
.container { display: flex; flex-wrap: wrap; gap: 24px; }
```

ITEM SIZING DYNAMICS

flex-grow, flex-shrink & flex-basis

EXPAND

flex-grow

Ability to absorb extra unused space. Default: 0.

CONTRACT

flex-shrink

Ability to shrink when space is tight. Default: 1.

START SIZE

flex-basis

Initial size before growing or shrinking occurs.

WEB DESIGN HOLY GRAIL

The Ultimate Centering Technique

The 3-Line Secret

Replaces 20 years of complex CSS centering hacks.

Full Coordinate Lock

Centers horizontally and vertically simultaneously.

```
/* Center any modal, card, or element */ .parent-container { display: flex; justify-content: center; /* Horizontal */ align-items: center; /* Vertical */ min-height: 100vh; }
```

REVIEW & IN-CLASS ACTIVITY

Session 10 Wrap-Up

Core Takeaways

- display: flex unlocks the power of 1D layout
- justify-content controls the main axis
- align-items controls the cross axis
- gap provides margin-free element spacing

Hands-On Lab Challenge

Build a modern Responsive Navigation Bar:

Features: Logo on the left, navigation links on the right using space-between, align-items: center, and 16px gap.